Pico 2 W HAT Schematic Design

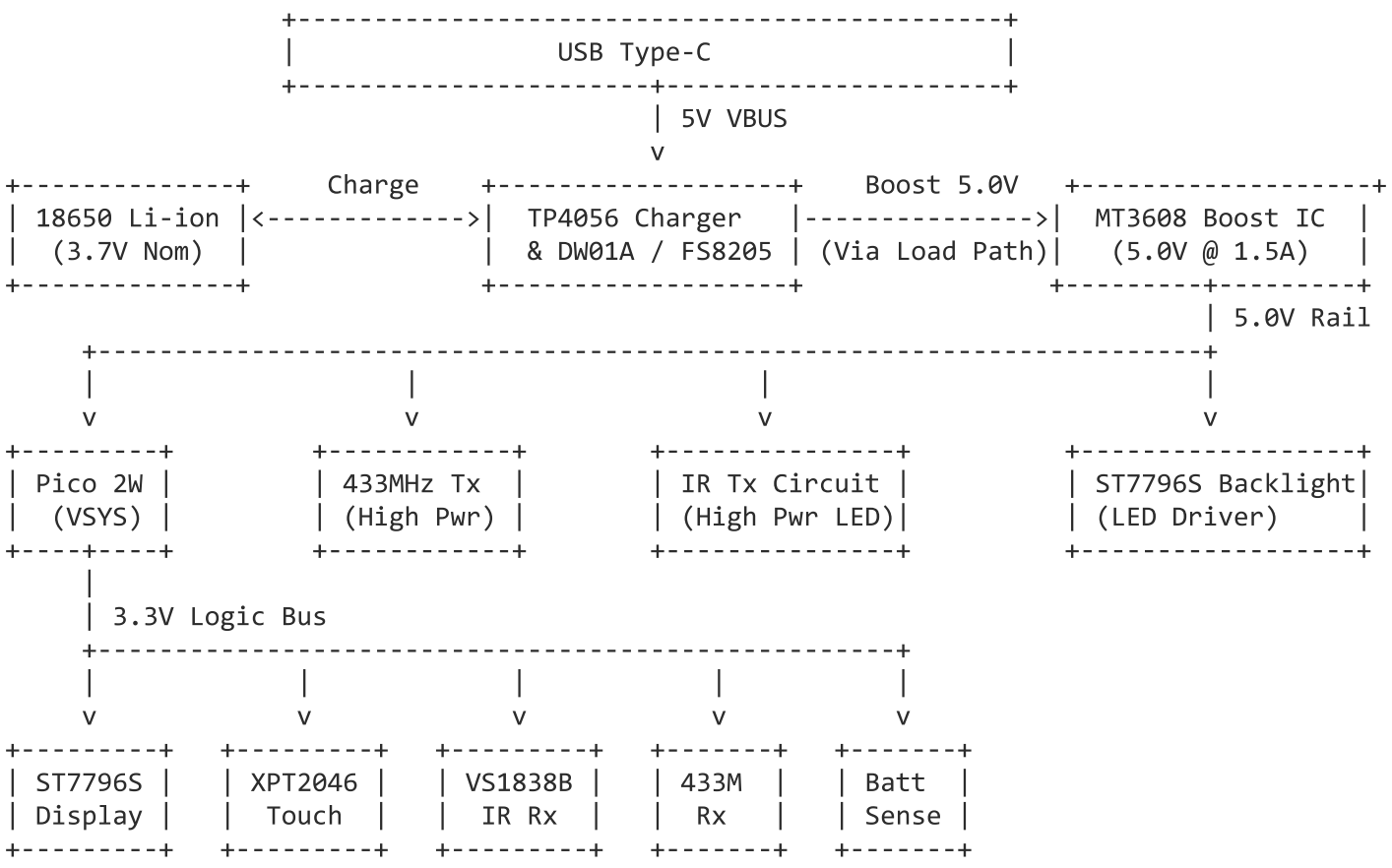
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User prompt: Make a schematic for a raspberry pi pico 2w hat with IR Receiver and Transmitter and a RF Receiver and Transmitter. And a ST7796S SPI Touchscreen. Powered by a 18650 battery with a charging circuit and a step up converter to feed the pico

Response: Here is a complete, production-ready schematic and hardware design guide for your **Raspberry Pi Pico 2 W HAT**.

This design features a fully autonomous power management system (18650 Li-ion charging, protection, automatic load sharing, and boost to 5V), an IR transceiver pair, an RF transceiver interface, and an SPI ST7796S display with an integrated XPT2046 touch controller.

1. System Block Diagram



2. Pico 2 W Master GPIO Mapping Table

The RP2350 microcontroller on the Pico 2 W has multiple dedicated hardware peripherals. The SPI0 bus is multiplexed between the Display and Touch controller using separate Chip-Select (CS) lines.

Pin Name	Pico Pin	Direction	Connected Device	Function / Notes
GPIO 12	Pin 16	Input	RF Receiver	Digital Data In (from RF Rx module)
GPIO 13	Pin 17	Output	RF Transmitter	Digital Data Out (to RF Tx module)

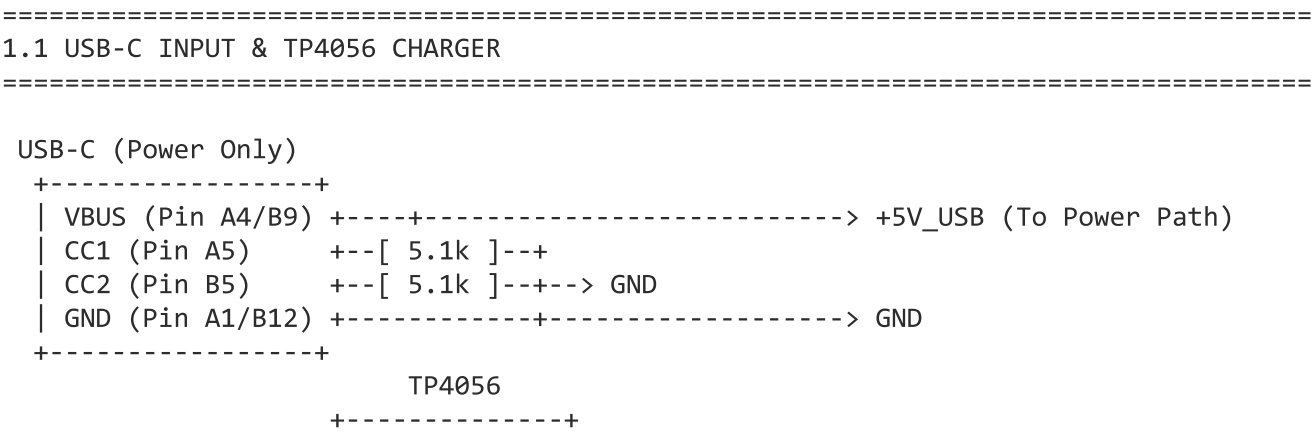
Pin Name	Pico Pin	Direction	Connected Device	Function / Notes
GPIO 14	Pin 19	Input	IR Receiver	Active-low Demodulated 38kHz input
GPIO 15	Pin 20	Output	IR Transmitter	38kHz PWM signal to MOSFET driver
GPIO 16	Pin 21	Input	ST7796S & Touch	**SPI0 MISO / SDO** (Shared)
GPIO 17	Pin 22	Output	ST7796S Display	**LCD_CS** (Chip Select, Active-Low)
GPIO 18	Pin 24	Output	ST7796S & Touch	**SPI0 SCK** (Clock, Shared)
GPIO 19	Pin 25	Output	ST7796S & Touch	**SPI0 MOSI / SDI** (Shared)
GPIO 20	Pin 26	Output	ST7796S Display	**LCD_DC** (Data/Command Select)
GPIO 21	Pin 27	Output	ST7796S Display	**LCD_RST** (Hardware Reset)
GPIO 22	Pin 29	Output	ST7796S Display	**LCD_BL** (Backlight PWM control)
GPIO 26	Pin 31	Output	XPT2046 Touch	**TOUCH_CS** (Chip Select, Active-Low)
GPIO 27	Pin 32	Input	XPT2046 Touch	**TOUCH_IRQ** (Pen-Down Interrupt)
GPIO 28	Pin 34	Analog In	Battery Sense	**ADC2** (Reads VBAT via 1:2 divider)
3V3_OUT	Pin 36	Power Out	System 3.3V Logic	Feeds logic of LCD, Touch, IR Rx, RF Rx
VSYS	Pin 39	Power In	Boost Converter	Receives **+5.0V** from MT3608
GND	Pins 38, etc.	Ground	System Ground	Common Return Path

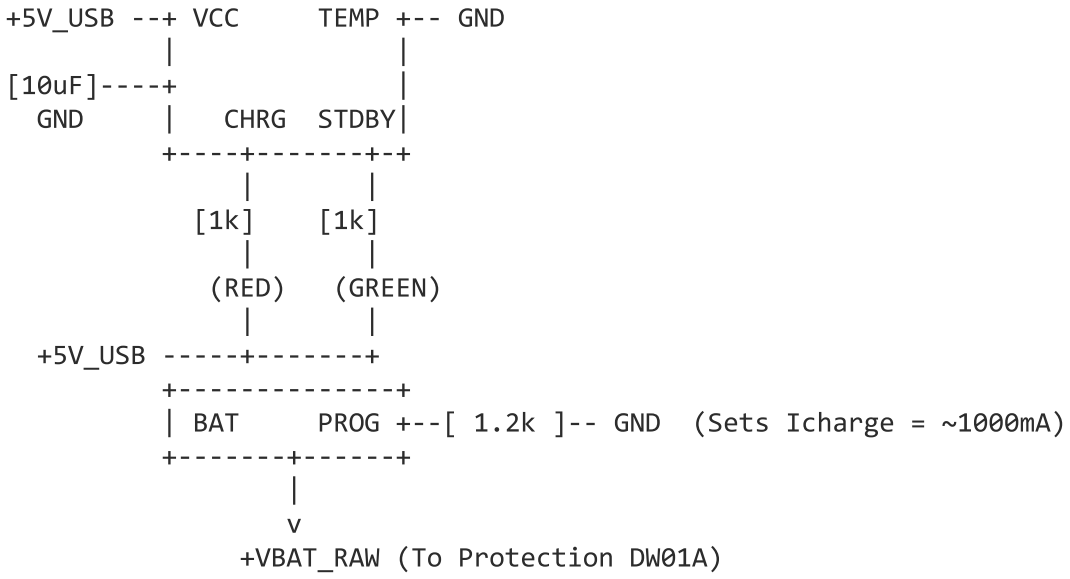
Remaining GPIOs (GPIO 0–11) are broken out onto an unpopulated header for future buttons, encoders, or expansion.

3. Detailed Circuit Schematics

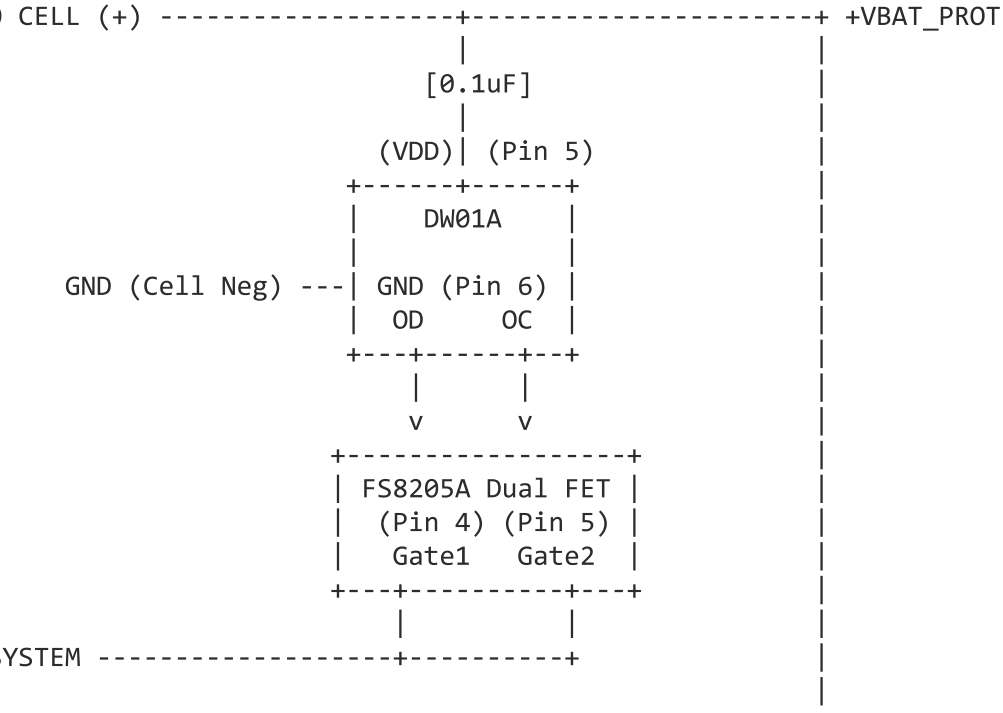
Module 1: Battery Management, Load Sharing & 5V Boost Converter

This section features **automatic power-path control**. When USB is plugged in, the system runs directly from USB while the TP4056 charges the cell. When USB is removed, the P-MOSFET turns on instantly with no reboot.

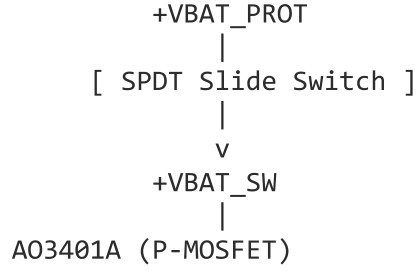


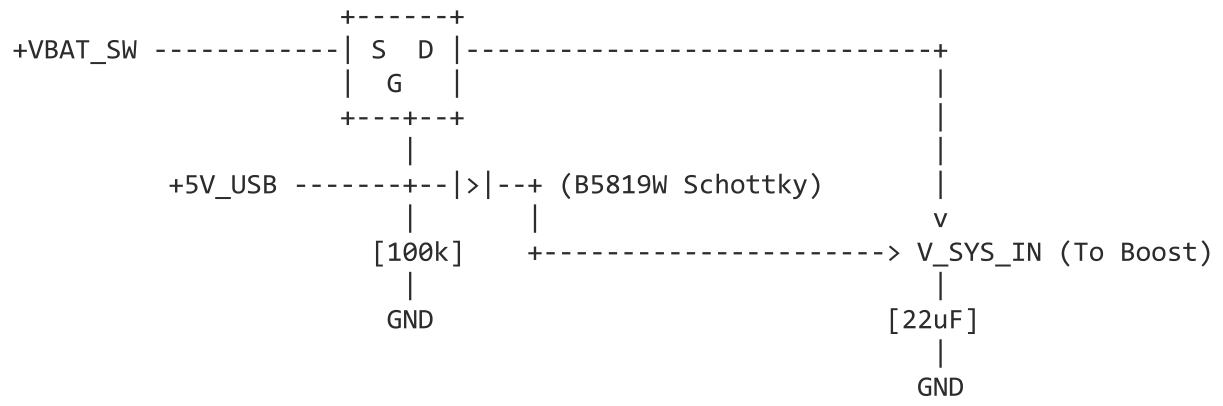


1.2 BATTERY PROTECTION CIRCUIT (DW01A + FS8205A)



1.3 POWER PATH SWITCH (LOAD SHARING) & SLIDE SWITCH






```
================================================================================
1.4 MT3608 STEP-UP CONVERTER (3.0V–4.2V -> 5.0V @ 1.5A)
================================================================================

                    10uH Inductor (>= 2.5A)
                      +---[ L1 ]---+
                      |            |
  V_SYS_IN ------------+            +------------+
  (From Path)          |        (SW)|            |
                     [22uF]     +----+----+     +-+-+  B5819W / SS34
                       GND      | MT3608  |     | |   Schottky Diode
                                |         |     +-+-+
             V_SYS_IN ----------+ EN   FB +-------+--------+------> +5V_SYS
                                |   GND   |       |        |        (To Pico VSYS,
                                +----+----+     [75k]   [22uF x2]    IR Tx, Display BL,
                                     |           |       GND        RF Tx)
                                    GND          +--[ 10k ]-- GND

  * Formula: Vout = 0.6V * (1 + 75k / 10k) = 5.1V (Compensates for trace drop)
```

Module 2: Battery Fuel Monitoring (ADC)

RP2350 ADC inputs tolerate a maximum of 3.3V. A high-value resistor divider scales the 4.2V maximum battery voltage to safe levels without causing parasitic drain.

```
                +VBAT_SW
                   |
                 [100k] (1% Tolerance)
                   |
                   +----------------------> GPIO 28 / ADC2 (Pin 34 on Pico 2W)
                   |
                 [100k] (1% Tolerance)
                   |
                  --- [0.1uF Ceramic filter]
                  ---
                   |
                  GND

* Voltage at GPIO 28 = VBAT / 2 (At 4.2V VBAT -> 2.10V ADC reading)
```

Module 3: Infrared (IR) Transceiver Subsystem

- **IR Receiver:** Standard 38kHz active receiver (VS1838B / TSOP38238). Powered from **3.3V** to keep logic lines directly compatible with the RP2350.
- **IR Transmitter:** 940nm high-power LED switched via an N-Channel logic-level MOSFET driven by GPIO 15 (PWM). Driven from the **5.0V** rail to maximize pulse power and range.

```
================================================================================
3.1 IR RECEIVER (VS1838B / TSOP38238)
================================================================================

       +3V3_OUT (Pico Pin 36)
           |
         [100R]
           |
           +---------------+-----[ 10k Pull-up ]----+
           |               |                       |
       +---+---+           |                       |
       |  VCC  |         [10uF]                    |
       |       |          GND                      v
       |  OUT  +---------------------------------> GPIO 14 (IR_RX)
       |       |
       |  GND  +---+
       +-------+   |
                  GND

================================================================================
3.2 IR TRANSMITTER (AO3400A / 2N7002 High-Pulse Driver)
================================================================================

      +5V_SYS
         |
      +--+--+
      |     | (Anode)
     (  IR  ) TSAL6200 / 940nm High Power IR LED
      |     | (Cathode)
      +--+--+
         |
       [ 15R ]  (1W Rating, limits peak pulse current to ~240mA)
         |
       +-+-+ Drain
       |   |
  +----+   | AO3400A (N-Channel Logic MOSFET)
  |    |   |
  |    +-+-+ Source
  |      |
  |     GND
  |
  +--[ 100R ]-----------------------------------> GPIO 15 (IR_TX / PWM)
  |
[100k] (Gate Pulldown to prevent floating HIGH at boot)
  |
 GND
```

Module 4: 433MHz / Sub-GHz RF Transceiver Subsystem

Designed for modular 433.92 MHz ASK/OOK modules (e.g., FS1000A / STX882 Transmitter and SYN480R / SRX882 Receiver) or a pin header footprint for a CC1101 transceiver board.

```
================================================================================
4.1 RF TRANSMITTER INTERFACE
================================================================================

             RF Transmitter Header (e.g., FS1000A / STX882)
             +-----------------------+
 +5V_SYS ----+ Pin 1: VCC            |
 GPIO 13 ----+ Pin 2: DATA           |   [ ANTENNA ]
             |                       |        |
             | Pin 3: ANT -----------+--------+ (17.3cm wire for 433MHz)
     GND ----+ Pin 4: GND            |
             +-----------------------+

================================================================================
4.2 RF RECEIVER INTERFACE (With 5V to 3.3V Logic Protection)
================================================================================

             RF Receiver Header (e.g., SYN480R / RXB6)
             +-----------------------+
 +5V_SYS ----+ Pin 1: VCC            |
             |                       |   [ ANTENNA ]
             | Pin 2: ANT -----------+--------+ (17.3cm wire)
             |                       |
     GND ----+ Pin 3: GND            |
             |                       |
             | Pin 4: DATA           +---+
             +-----------------------+   |
                                       [ 1k ]
                                         |
 GPIO 12 (RF_RX) <-----------------------+
                                         |
                                       [ 2k ]
                                         |
                                        GND
  *(1k / 2k voltage divider steps down a 5V receiver logic signal to 3.3V safe for Pico)*
```

Module 5: ST7796S SPI Display & Resistive Touch (XPT2046)

Standard 3.5" or 4.0" ST7796S SPI displays have a unified single-row connector containing pins for the TFT and the Touch controller.

```
================================================================================
5.1 DISPLAY & TOUCH PIN INTERCONNECT
================================================================================

 ST7796S + XPT2046 Breakout Header               Pico 2 W Connection
+-----------------------------------+          +-----------------------------------+
| Pin  1: VCC                      +<---------+ +5V_SYS (Or +3V3_OUT per board)  |
| Pin  2: GND                      +--------->+ GND                               |
| Pin  3: LCD_CS (Chip Select)     +<---------+ GPIO 17                           |
| Pin  4: LCD_RST (Reset)          +<---------+ GPIO 21                           |
| Pin  5: LCD_DC (Command/Data)    +<---------+ GPIO 20                           |
| Pin  6: SDI / MOSI (LCD Data In) +<---------+ GPIO 19 (SPI0_TX)                 |
```

```
| Pin  7: SCK (SPI Clock)            +<----------+ GPIO 18 (SPI0_SCK)                |
| Pin  8: LED / BLK (Backlight)      +<----------+ To Backlight Driver (See below)   |
| Pin  9: SDO / MISO (LCD Data Out)  +---------->+ GPIO 16 (SPI0_RX)                  |
| Pin 10: T_CLK (Touch Clock)        +<----------+ GPIO 18 (SPI0_SCK)  [Shared SPI] |
| Pin 11: T_CS (Touch Chip Select)   +<----------+ GPIO 26                           |
| Pin 12: T_DIN (Touch Data In)      +<----------+ GPIO 19 (SPI0_TX)   [Shared SPI] |
| Pin 13: T_DO (Touch Data Out)      +---------->+ GPIO 16 (SPI0_RX)   [Shared SPI] |
| Pin 14: T_IRQ (Touch Interrupt)    +---------->+ GPIO 27                           |
+-----------------------------------+          +---------------------------------+

================================================================================
5.2 BACKLIGHT DIMMING DRIVER (PWM)
================================================================================

      +5V_SYS (Or 3.3V per screen spec)
         |
         +-----[ LCD Pin 8 (LED / BLK) ]
                      |
                     +-+-+ Drain
                     |   |
  GPIO 22 ---------+ |   | AO3400A / 2N7002
  (LCD_BL)         | |   |
                   +-+-+ Source
                     |
                    GND
```

4. Complete Bill of Materials (BOM)

Reference	Value / Part Number	Description	Recommended Package
U1	Raspberry Pi Pico 2 W	Dual ARM Cortex-M33 / RISC-V with Wi-Fi/BLE	DIP-40 Module / SMD Castile
U2	TP4056	1A Standalone Linear Li-ion Battery Charger	SOP-8-PP
U3	DW01A	Battery Protection IC	SOT-23-6
U4	FS8205A	Dual N-Channel 20V 4A MOSFET (for DW01A)	TSSOP-8
U5	MT3608	2A High-Efficiency Boost Converter	SOT-23-6
Q1	AO3401A	P-Channel MOSFET (-30V, -4A) for load sharing	SOT-23-3
Q2, Q3	AO3400A	N-Channel MOSFET (30V, 5A) for IR Tx & Backlight	SOT-23-3
D1, D2	B5819W / SS34	Schottky Barrier Diode (40V, 1A–3A)	SOD-123 / SMA
D3	TSAL6200	940nm High Power Infrared Emitting Diode	5mm Through-Hole

Reference	Value / Part Number	Description	Recommended Package
IR1	VS1838B / TSOP38238	38kHz IR Receiver Module	Through-Hole 3-pin
DISP1	ST7796S 3.5" or 4.0"	SPI TFT LCD (480x320) with XPT2046 Touch	14-Pin 2.54mm Header
RF1, RF2	433MHz Tx/Rx Pair	FS1000A / STX882 (Tx) + SYN480R / SRX882 (Rx)	2.54mm Headers
L1	10µH (≥ 2.5A)	Shielded Power Inductor	SMD 5x5mm (CD54 or NR5040)
J1	TYPE-C-31-M-12	16-Pin / 6-Pin Power-only USB Type-C Receptacle	SMD
BT1	18650 Battery Clip	Single 18650 SMD / Through-hole battery holder	Keystone 1042 / 1048
SW1	SPDT Slide Switch	Power ON/OFF switch (Sub-miniature)	SMD / 2.54mm THT
R1, R2	5.1kΩ (1%)	USB-C CC1/CC2 pulldowns	0603
R3	1.2kΩ (1%)	TP4056 PROG resistor (sets 1A current)	0603
R4, R5	1kΩ	LED Current limiters for CHRG/STDBY	0603
R6	100kΩ	Power Path gate pull-down	0603
R7	75kΩ (0.5% or 1%)	MT3608 Feedback divider (R_top)	0603
R8	10kΩ (1%)	MT3608 Feedback divider (R_bottom)	0603
R9, R10	100kΩ (1%)	Battery Sense voltage divider	0603
R11	15Ω (1W)	IR LED Current limiter resistor	1206 / 2512
R12	100Ω	IR Receiver supply filtering resistor	0603
R13, R14	1kΩ, 2kΩ	RF Receiver 5V-to-3.3V Logic level divider	0603
C1, C2	10µF (16V)	Input/Output Filter Caps for TP4056	0805 X5R
C3, C4	22µF (16V)	Boost Converter Input & Output MLCC Caps	0805 / 1206 X5R
C5, C6	0.1µF (50V)	Decoupling capacitors (DW01A, ADC filter)	0603

5. PCB Layout and Routing Guidelines

1. **Boost Converter Switching Loop (Critical):**
    - Keep the loop containing **L1 -> D2 (Schottky) -> C4 (Output Cap) -> MT3608 GND** as physically small and tight as possible.

- Wide copper pours (at least 1.0mm) should be used for the `SW` pin node to handle the high switching frequency (1.2MHz) without generating EMI.

2. **RF Antennas:**
    - Keep both the 433MHz receiver and transmitter trace runs far away from the MT3608 inductor and switching diode.
    - Provide an unshielded edge pad with a hole for a 17.3cm wire whip antenna (or a SMA connector footprint) with a **ground plane cut-out (keep-out)** directly under the antenna solder landing.
3. **SPI Trace Matching:**
    - The ST7796S display can run SPI clock frequencies up to 40MHz–60MHz. Route `SCK`, `MOSI`, and `MISO` with equal trace lengths and avoid routing them directly parallel to the 433MHz data line or the high-current IR LED drive path.
4. **Thermal Considerations:**
    - The **TP4056** dissipates heat during constant-current mode (up to 1W). Place 4 to 6 ground vias directly inside and around its exposed thermal pad to sink heat into the bottom ground plane.
    - Make sure the battery holder terminals have beefy traces (>1.5mm) to minimize resistance drops under peak loads.
